Math 215, Second Midterm Exam, November 10, 2022, 6:00–7:30 pm

Instructions

1. **Do not open this exam until you are told to do so**.
2. This examination booklet contains 8 problems on 11 sheets of paper including the front cover. The second to last sheet is left blank for scratch work. The last sheet contains a list of formulas.
3. Do not separate the pages of this exam, other than the formula sheet at the end of the exam. Separated pages will not be graded.
4. Please read the instructions for each individual problem carefully. One of the skills being tested on this exam is your ability to interpret mathematical questions, so instructors will not answer questions about exam problems during the exam.
5. Show an appropriate amount of work (including appropriate explanation) for each problem, so that graders can see not only your answer but how you obtained it. Still, you should **mark your answers in the correct box**. Include units in your answer where that is appropriate.
6. If needed, you may continue your solution on the back side of a page.
7. This is a closed book exam. Electronic devices, calculators and note-cards are not allowed. **Turn off all cell phones, remove all headphones, and place any watch you are using on the desk in front of you**.

Name:

UMID:

Section:

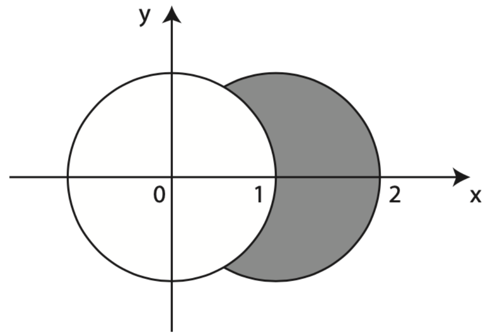
(The sections meet at as follows: Section 10: 8-9 am, Section 30: 9-10 am, Section 40: 10-11 am, Section 50: 11am-12 pm, Section 60: 12-1 pm, Section 70: 1-2 pm, Section 80: 2-3 pm, Section 90: 3-4 pm)

Point distribution of the problems on the exam

Question:	1	2	3	4	5	6	7	8	Total
Points:	12	13	10	14	10	14	14	13	100

This page will not be graded

1. (12 points) Let D be the region in the plane outside the circle $x^2+y^2=1$ and inside the circle $x^2+y^2=2x$, see the shaded part in the picture below. Consider an inhomogeneous lamina whose shape is D, and where the density is inversely proportional to the distance to the origin, that is, $\rho(x,y)=c/\sqrt{x^2+y^2}$, in such a way that the largest density on D is 1. Find the mass of the lamina.



mass:

Extra writing area for Question 1

2. Let $f(x, y) = xye^{x^2 - y^5}$.

(a) (4 points) Compute $\nabla f(1, 1)$

$\nabla f(1, 1)$:

(b) (3 points) Either find a unit vector $\mathbf{u}$ at $(1, 1)$ such that $D_{\mathbf{u}} f(1, 1) = -5$, or explain why such a unit vector $\mathbf{u}$ does not exist. Please show work.

$\mathbf{u}$:

(c) (3 points) Either find a unit vector $\mathbf{u}$ at $(1, 1)$ such that $D_{\mathbf{u}} f(1, 1) = 0$, or explain why such a unit vector $\mathbf{u}$ does not exist. Please show work.

$\mathbf{u}$:

(d) (3 points) Either find a unit vector $\mathbf{u}$ at $(1, 1)$ such that $D_{\mathbf{u}} f(1, 1) = 2\pi$, or explain why such a unit vector $\mathbf{u}$ does not exist. Please show work.

$\mathbf{u}$:

Extra writing area for Question 2

3. Check the box (like this ☑) for each statement that is true. Leave the box blank for each statement that is false. No justification needed.

(a) (2 points) ☐

If two smooth functions $f(x, y)$ and $g(x, y)$ each have a saddle point at $(0, 0)$, so must the function $f(x, y) + g(x, y)$.

(b) (2 points) ☐

If $D = \{(x, y) \mid -1 \leq x + y \leq 1, -1 \leq x - y \leq 1\}$, then, for any continuous function $f(x, y)$,

$$\iint_D f(x, y)\, dA = \int_{-1}^{1} \int_{-1}^{1} f(x + y, x - y)\, dy dx.$$

(c) (2 points) ☐

Let C be the ellipse cut out by the two equations $z = 2 - x^2 - \frac{1}{2}y^2$ and $z = x + y$. If P is the point on C at which the function $f(x, y, z) = x^2 + y^2 + z^2$ takes its maximum value, then the level surface of f passing though P is perpendicular to C at P.

(d) (2 points) ☐

For all continuous functions $f(x)$ we have $\int_0^1 \int_0^1 f(x)\, dy dx = \int_0^1 f(x)\, dx$.

(e) (2 points) ☐

If $f(x, y, z)$ is a smooth function on $\mathbb{R}^3$, then there exists a unit vector $\mathbf{u}$ such that

$$D_{\mathbf{u}} f(1, 2, 3) = -\frac{\partial f}{\partial x}(1, 2, 3).$$

Extra writing area for Question 3

4. Consider the iterated integral

$$I = \int_{-1}^{1} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \int_{1}^{\sqrt{2-x^2-y^2}} \frac{6x^2 z}{(x^2+y^2)^{3/2}}\, dzdydx.$$

(a) (6 points) Write (but do not evaluate) I in cylindrical coordinates.

I:

(b) (8 points) Evaluate I.

I:

Extra writing area for Question 4

5. Check the box (like this ☑) for each statement that is true. Leave the box blank for each statement that is false. No justification needed.

(a) (2 points) ☐

The surface area of the graph $z = \frac{1}{1+x^{2022}+y^{2022}}$ over the disc $x^2 + y^2 \leq 1$ is equal to $\pi/2$.

(b) (2 points) ☐

The maximum value of the function $xe^{-(x^2+y^2)^3}$ on the circle $x^2 + y^2 = 1$ is equal to 1.

(c) (2 points) ☐

If a liquid fills up the unit ball $x^2 + y^2 + z^2 \leq 1$ and has density $\rho(x, y, z) = 3 + xy^3z^5$, then the mass of the liquid is 4π.

(d) (2 points) ☐

If the function $f(x, y)$ is given by the table below, then, based on midpoint rule with $m = n = 1$, we have that $\int_1^3 \int_{-2}^0 f(x, y)\, dydx > \int_0^2 \int_3^5 f(x, y)\, dxdy$.

x\y	-2	-1	0	1	2
1	14	2	8	15	-3
2	5	9	1	14	-1
3	5	7	-1	8	4
4	16	15	0	5	8
5	13	0	1	1	14

(e) (2 points) ☐

For any continuous function $f(x, y)$, we have

$$\int_0^1 \int_0^{x^2} f(x, y)\, dydx = \int_0^1 \int_{y^2}^1 f(x, y)\, dxdy.$$

Extra writing area for Question 5

6. (14 points) Let E be the part of the first octant (i.e. $x \geq 0$, $y \geq 0$, $z \geq 0$) that lies outside the sphere of radius 1 and inside the sphere of radius 2. Compute the triple integral

$$I = \iiint_E \frac{3}{1 + (x^2 + y^2 + z^2)^{3/2}} \, dV.$$

I: $\boxed{}$

Extra writing area for Question 6

7. (14 points) Find the absolute maximum value and the absolute minimum value of the function $f(x, y) = 2xy - x - y$ on the (solid) triangle with vertices at $(0, 0)$, $(4, 0)$, and $(0, 4)$.

Absolute maximum value: $\boxed{}$ Absolute minimum value: $\boxed{}$

Extra writing area for Question 7

8. Let F be the region in space above the xy-plane and below the parabolic cylinder $z = 1 - x^2$. Let E be the part of F where $x \geq 0$ and $|y| \leq 2x$.

(a) (2 points) Sketch (roughly) E and F.

(b) (3 points) Write $I = \iiint_E f(x, y, z)\, dV$ as an iterated integral $\int_{a_1}^{a_2} \int_{g_1(x)}^{g_2(x)} \int_{h_1(x,y)}^{h_2(x,y)} f(x, y, z)\, dzdydx$, where you need to determine a_1, a_2, $g_1(x)$, $g_2(x)$, $h_1(x, y)$, and $h_2(x, y)$. **Please read the labels carefully and don't mix up the boxes!**

a_1:		$g_1(x)$:		$h_1(x, y)$:	
a_2:		$g_2(x)$:		$h_2(x, y)$:	

(c) (4 points) Compute the volume of E. Please show work.

$V(E)$:

(d) (4 points) Compute the integral $I = \iiint_E (2 + \sin(y))\, dV$. Please show work.

I:

Extra writing area for Question 8

Page for scratch work. Not graded if torn off.

Page for scratch work. Not graded if torn off.

Formula sheet. Not graded. May be torn off.

(Not all of these formulas are necessarily needed for this exam.)

- Integration by parts: $\int u dv = uv - \int v du$.
- $\sin^2(x) + \cos^2(x) = 1$, $\cos(2x) = \cos^2(x) - \sin^2(x)$, $\sin(2x) = 2\sin(x)\cos(x)$
- $\sin^2(x) = \dfrac{1-\cos(2x)}{2}$, $\cos^2(x) = \dfrac{1+\cos(2x)}{2}$
- $\sin x \sin y = \frac{1}{2}(\cos(x-y) - \cos(x+y))$, $\cos x \cos y = \frac{1}{2}(\cos(x-y) + \cos(x+y))$.
- $\sin x \cos y = \frac{1}{2}(\sin(x+y) + \sin(x-y))$
- $\cos(\pi/3) = 1/2$, $\sin(\pi/3) = \sqrt{3}/2$, $\cos(\pi/4) = \sqrt{2}/2$, $\sin(\pi/4) = \sqrt{2}/2$, $\cos(\pi/6) = \sqrt{3}/2$, $\sin(\pi/6) = 1/2$, $\cos(0) = 1$, $\sin(0) = 0$.
- $\frac{d}{dx}\sin(x) = \cos(x)$, $\frac{d}{dx}\cos(x) = -\sin(x)$.
- Polar coordinates $x = r\cos(\theta)$, $y = r\sin(\theta)$.
- Cylindrical coordinates $x = r\cos(\theta)$, $y = r\sin(\theta)$, $z = z$.
- Spherical coordinates $x = \rho\cos(\theta)\sin(\phi)$, $y = \rho\sin(\theta)\sin(\phi)$, $z = \rho\cos(\phi)$.
- Surface area of the graph of the function $f(x,y)$ over a region D is $\iint_D \sqrt{1 + f_x^2 + f_y^2}\, dA$.
- The circumference of a circle of radius a is $2\pi a$.
- The area of a disc of radius a is πa^2.
- The volume of a right circular cylinder of radius a and height h is $\pi a^2 h$.
- The volume of a sphere of radius a is $4\pi a^3/3$.
- The surface area of a sphere of radius a is $4\pi a^2$.
- The volume of a cone with base radius a and height b is $\frac{1}{3}\pi a^2 b$.

This page will not be graded. May be torn off.